Integrated Circuits vs Discrete-Components Circuits

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1 The Market

In order to understand which kind of electronics is the more important for companies nowadays, the first step is to analyze the values in terms of market. In most of the reports we can find that the biggest part of semiconductor devices market is made by the integrated part. The World Semiconductor Trade Statistics (WSTS) report that in 2024, the total market of semiconductor values 630,549 US$M, and the integrated-circuit part values 539,505 US$M, approximately the 85 % of all the market. While discrete semiconductors stand only at 31,026 US$M, which is the 5 % [8].

Autumn 2025	Amounts in US$M			Year on Year Growth in %		
	2024	2025	2026	2024	2025	2026
Americas	195,123	**251,926**	**338,574**	45.2	29.1	34.4
Europe	51,250	**54,127**	**60,429**	-8.1	5.6	11.6
Japan	46,739	**44,835**	**50,164**	0.0	-4.1	11.9
Asia Pacific	337,437	**421,354**	**526,293**	16.4	24.9	24.9
Total World - $M	630,549	**772,243**	**975,460**	19.7	22.5	26.3
Discrete Semiconductors	31,026	**30,900**	**33,436**	-12.7	-0.4	8.2
Optoelectronics	41,095	**42,597**	**45,020**	-4.8	3.7	5.7
Sensors	18,923	**20,894**	**22,713**	-4.1	10.4	8.7
Integrated Circuits	539,505	**677,852**	**874,291**	25.9	25.6	29.0
Analog	79,588	85,552	91,988	-2.0	7.5	7.5
Micro	78,633	84,839	96,620	3.0	7.9	13.9
Logic	215,768	295,892	390,863	20.8	37.1	32.1
Memory	165,516	211,568	294,821	79.3	27.8	39.4
Total Products - $M	630,549	**772,243**	**975,460**	19.7	22.5	26.3

FIGURE 1: WSTS Forecast Summary

2 The Companies

In order to analyze which of the two is more important we can also check the big companies and see in which sector are they working and how big they are (in both terms of revenue and number of employees). A summary is shown in table 1, that groups some of the biggest electronic companies.

Company	Revenue	Employees
NVIDIA	130.5 $B	36,000
Intel	52.9 $B	85,100
AMD	34.6 $B	31,000
Infineon	14.7 €B	57,000
Texas Instruments	17.7 $B	33,000
Analog Devices	11.0 $B	24,500
STMicroelectronics	11.8 $B	49,157

TABLE 1: Revenue and number of employees of selected semiconductor companies in 2025

2.1 IC Companies

In the table shown, almost all the companies are mostly IC-based.

The first company in terms of revenue is NVIDIA, which sells IC devices as CPUs, GPUs and DPUs. [5]

Intel is the one with the most number of employees, and it produces Microprocessors as well as SoCs, GPUs, NPUs, ASICSs and more. [4]

AMD is another fully IC company that in terms of business is similar to NVIDIA, selling CPUs, APUs and SoC. [1]

Texas Instruments normally sells Analog and Embedded Processing devices based on IC as well as Analog Devices that sells various IC devices in general for Mixed Signal purposes. [7] [2]

2.2 Mixed Companies

There are a lot of companies that sell both IC devices and Discrete devices. The first one that in the table is Infineon, from the financial report of 2025 we can estimate that approximatly 65% of the revenues of the company comes from IC, while the discrete part takes around 30%. [3]

Also STMicroelectronics sells both the type of devices, having approximatly 10% of discrete devices and 70% of IC devices. [6]

2.3 Discrete Circuits Companies

Almost all the comanies work at least a part with IC devices, some exeptions are Nexperia and Vishay for example, both focused on devices as MOSFETs, Diodes, BJTs and more.

3 Conclusion

In conlusion, considering the data both from the market and from the single companies we can say that the majority of the elcectronic market is based on IC devices.

On the company side, we can surely say that given the big amount of emplyees and the large amount of IC devices, the request of IC expert is surely higher than the discrete one.

References

[1] Advanced Micro Devices, Inc. 2025 annual report. Technical report, Advanced Micro Devices, Inc., 2026. Fiscal year ended December 27, 2025. Revenue: $34.639 billion; 31,359 employees.

[2] Analog Devices, Inc. 2025 annual report. Technical report, Analog Devices, Inc., 2025. Fiscal year ended November 1, 2025. Revenue: $11.020 billion; approximately 24,500 employees.

[3] Infineon Technologies AG. Annual report 2025. Technical report, Infineon Technologies AG, 2025. Fiscal year ended September 30, 2025. Revenue: approximately EUR 14.7 billion; approximately 57,000 employees.

[4] Intel Corporation. 2025 annual report. Technical report, Intel Corporation, 2026. Fiscal year ended December 27, 2025. Revenue: $52.853 billion; 85,100 employees.

[5] NVIDIA Corporation. Annual report 2025. Technical report, NVIDIA Corporation, 2025. Fiscal year ended January 26, 2025. Revenue: $130.497 billion; approximately 36,000 employees.

[6] STMicroelectronics N.V. 2025 annual report. Technical report, STMicroelectronics N.V., 2026. Fiscal year ended December 31, 2025. Net revenues: $11.800 billion; 49,157 employees.

[7] Texas Instruments Incorporated. 2025 annual report. Technical report, Texas Instruments Incorporated, 2026. Fiscal year ended December 31, 2025. Revenue: $17.682 billion; approximately 33,000 employees.

[8] World Semiconductor Trade Statistics. Global semiconductor market approaches USD 1 trillion in 2026. Autumn 2025 forecast release.